

Assembly of the two-loop MHV integrand from maximally positive mutual numerators at five and six points

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Abstract

For the two-loop four-point MHV amplituhedron, the canonical form on each positive chart is known to be a pure dlog form times the ratio D^+/D , where $D = \langle ABCD \rangle$ is the mutual positivity condition and D^+ is the sum of its positive monomials [1]. We test, by exact computation, whether this “maximally positive numerator” structure *assembles* at higher multiplicity: whether the full two-loop MHV integrand at $n = 5, 6$ equals the sum over all composite cells (ordered pairs of one-loop sign-flip cells) of $\text{Kermit} \times \text{Kermit} \times D^+/D$, with no corrections. It does. At $n = 5$ (9 composite cells) and $n = 6$ (36 composite cells) the assembly reproduces the local double-pentagon representation of the integrand [2] with ratio exactly 1 in rational arithmetic at independent generic kinematic points; interior seams cancel pairwise; and all well-defined transverse octa-cut residues at $n = 5$ are quantized as required. Two structural observations explain why the preconditions cannot fail at any n : in the sign-flip charts, $\langle ABCD \rangle$ is automatically multilinear in the eight chart variables, and every monomial coefficient is a single Plücker bracket, so the sign split defining D^+ is independent of the choice of positive kinematics. In particular the quasi-linear complication anticipated in [1] does not arise for two-loop MHV in these raw charts at $n \leq 6$, suggesting the naive assembly may hold for all n . All computations are exact and reproducible from the public repository.

1 Setup

We use momentum twistors $Z_1, \dots, Z_n$ on a moment curve (all ordered Plücker brackets positive) and the one-loop MHV amplituhedron in its sign-flip description [1, 3, 4]: a loop line AB with $\langle AB i i+1 \rangle > 0$ (twisted cyclic) and exactly two sign flips in the sequence $\langle AB 1 i \rangle$, $i = 2, \dots, n$. The flip positions (i, j) , $2 \leq i < j \leq n-1$, label $(n-2)(n-3)/2$ cells, each with the positive chart

$$A = Z_1 + x Z_i + w Z_{i+1}, \quad B = -Z_1 + y Z_j + z Z_{j+1}, \quad x, w, y, z > 0. \quad (1)$$

On each cell the canonical form is the corresponding “Kermit” term [2],

$$K_{ij}(A, B) = \frac{\langle AB (1 i i+1) \cap (1 j j+1) \rangle^2}{\langle AB 1 i \rangle \langle AB i i+1 \rangle \langle AB 1 i+1 \rangle \langle AB 1 j \rangle \langle AB j j+1 \rangle \langle AB 1 j+1 \rangle}, \quad (2)$$

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which we verify to be an exact dlog form on its chart ($K_{ij} \langle AB \ i \ i+1 \rangle \langle AB \ j \ j+1 \rangle x y z = 1$ identically) for all cells at $n = 5, 6$.

At two loops, the geometry is the set of ordered pairs of lines (AB, CD) , each in the one-loop region, with the single mutual condition $\langle ABCD \rangle > 0$ [1, 4, 5]. Composite cells are ordered pairs of one-loop cells; in each composite chart write $D = \langle ABCD \rangle$ as a polynomial in the eight chart variables and let D^+ denote the sum of its positive-coefficient monomials.

Two structural facts. (i) D is automatically multilinear: each chart variable enters exactly one column of the 4×4 determinant. (ii) Each monomial coefficient is a single Plücker bracket $\pm \langle Z_a Z_b Z_c Z_d \rangle$ (one basis vector chosen per column), so its sign is fixed by the ordering of the moment curve alone: the split $D = D^+ - D^-$ is independent of the choice of positive Z 's. We verified both facts computationally for all 9 composite cells at $n = 5$ and all 36 at $n = 6$ (monomial counts ranging from 8 to 26), in addition to their one-line proofs. Consequently the “maximally positive numerator” is well defined at every n , and the only question is whether the assembly reproduces the integrand.

2 The assembly and its tests

Postulate. With all quantities evaluated at common representatives of the loop lines,

$$M_2^{(n)} = \sum_{\text{composite cells}} K_{c_1}(A, B) K_{c_2}(C, D) \frac{D_{c_1 c_2}^+}{D_{c_1 c_2}}, \quad (3)$$

where $D_{c_1 c_2}$, $D_{c_1 c_2}^+$ are evaluated by canonically extracting each pair's chart coordinates from the lines, and the sum is over *ordered* pairs (so $M_2 = 2 \mathcal{A}^{2\text{-loop}}$ in the convention where the physical amplitude carries 1/2 for indistinguishable loops). At $n = 4$ this is the known result [1], which our implementation reproduces exactly.

Gold standard. We compare (3) against the local double-pentagon representation [2], $\mathcal{A}_{\text{MHV}}^{2\text{-loop}} = \frac{1}{2} \sum_{i < j < k < l < i} I_{ijkl}$, with the chiral numerators $\langle AB \ (i-1 \ i \ i+1) \cap (j-1 \ j \ j+1) \rangle \langle ijkl \rangle \langle CD \ (k-1 \ k \ k+1) \cap (l-1 \ l \ l+1) \rangle$, implemented verbatim from the reference. In exact rational arithmetic at independent generic kinematic points:

n	composite cells	double pentagons	ratio $M_2 / \sum I$
5	9	20	1 (three points)
6	36	60	1 (two points)

The ratios are exact rational numbers, not floating-point agreements.

Seams. Individual composite-cell terms have poles on the spurious hyperplanes $\langle AB1i \rangle$ (i non-adjacent). Along exact one-parameter rational paths onto each seam at $n = 5$, individual terms diverge as $1/\varepsilon$ while the assembled sum is constant to six digits across four orders of magnitude in ε , and the sum diverges as required on the physical mutual pole $\langle ABCD \rangle \rightarrow 0$.

Leading singularities ($n = 5$). On the two standard octa-cut families with rational Schubert solutions, all well-defined transverse residues of the assembly are exactly 0, in the two ways the physics demands: 50/50 zeros on kissing-pentagon cuts (at $n = 5$ all disjoint non-adjacent chords cross, so every transverse kissing cut is non-planar), and 20/20 zeros on the plane-intersection branches of pentagon-box cuts (where the chiral numerators are designed to vanish). The remaining chord-branch pentagon-box configurations at $n = 5$ are composite strata (cut solutions pass through polygon vertices; naive transverse residues are ray-dependent, which our procedure detects automatically); their iterated-residue census is left open. The residue machinery is calibrated in-code on the one-loop box quadruple cut (residues exactly ∓ 1 , invariant under representative changes).

3 Discussion

Three comments. First, the result is a verification with an unexpected edge: the quasi-linear machinery of [1], developed because “the tedious triangulation is inevitable” in general, is never needed for two-loop MHV in the raw sign-flip charts at $n \leq 6$ — the naive term-by-term positive part assembles into the exact integrand. Given the structural facts (i)–(ii), which hold at every n , it is natural to conjecture that (3) holds for all n at two loops MHV; we have no proof, and the composite leading-singularity sector remains unchecked beyond the families listed. Second, two predictions made during this work were refuted by its own tests and are recorded in the repository alongside the confirmations: an expectation of unit transverse residues on $n = 5$ kissing cuts (planarity forces zeros instead), and the expectation that quasi-linearity would first appear at $n = 5$ or 6 (it does not). Third, everything here is exact, small, and reproducible: the computations run in minutes on a laptop from the public repository, and we would welcome correction or extension by practitioners — in particular a proof or counterexample for the all- n conjecture, and the composite-residue census.

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References

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